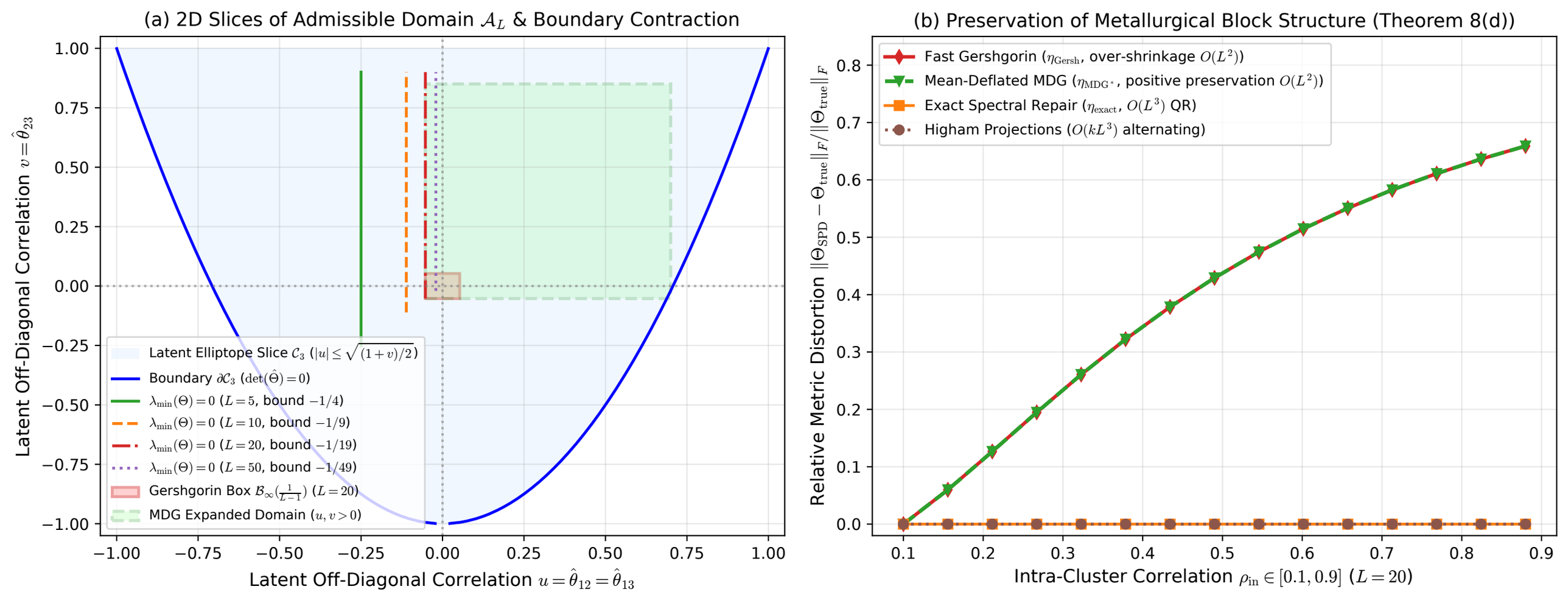
(a) 2D Slices of Admissible Domain $\mathcal{A}_L$ & Boundary Contraction
Latent Off-Diagonal Correlation $v = \hat{\theta}_{23}$
Latent Off-Diagonal Correlation $u = \hat{\theta}_{12} = \hat{\theta}_{13}$
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Latent Elliptope Slice $\mathcal{C}_3$ ($|u| \leq \sqrt{(1+v)/2}$)
Boundary $\partial\mathcal{C}_3$ ($\det(\hat{\Theta}) = 0$)
$\lambda_{\min}(\Theta) = 0$ ($L = 5$, bound $-1/4$)
$\lambda_{\min}(\Theta) = 0$ ($L = 10$, bound $-1/9$)
$\lambda_{\min}(\Theta) = 0$ ($L = 20$, bound $-1/19$)
$\lambda_{\min}(\Theta) = 0$ ($L = 50$, bound $-1/49$)
Gershgorin Box $\mathcal{B}_\infty(\frac{1}{L-1})$ ($L = 20$)
MDG Expanded Domain ($u, v > 0$)
(b) Preservation of Metallurgical Block Structure (Theorem 8(d))
Relative Metric Distortion $\|\Theta_{\text{SPD}} - \Theta_{\text{true}}\|_F / \|\Theta_{\text{true}}\|_F$
Intra-Cluster Correlation $\rho_{\text{in}} \in [0.1, 0.9]$ ($L = 20$)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Fast Gershgorin (η_{Gersh}, over-shrinkage $O(L^2)$)
Mean-Deflated MDG (η_{MDG^*}, positive preservation $O(L^2)$)
Exact Spectral Repair (η_{exact}, $O(L^3)$ QR)
Higham Projections ($O(kL^3)$ alternating)